**Project Design Phase-II**
**Solution Requirements (Functional & Non-functional)**

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|---|---|
| Date | 30 September 2026 |
| Team ID | SWTID-2026-2365 |
| Project Name | AI FitTrack API – Fitness Tracking API with Google Gemini AI |
| Maximum Marks | 4 Marks |

**Functional Requirements:**

Following are the functional requirements of the proposed solution.

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task) |
|---|---|---|
| FR-1 | User Registration | Register with name, email and password<br>Duplicate email check<br>Password stored as bcrypt hash<br>JWT token returned after registration |
| FR-2 | User Authentication | Login with email and password<br>JWT token (30-day validity) issued on login<br>View profile using Bearer token |
| FR-3 | Workout Management | Add workout (name, category, duration, calories, date)<br>View all workouts, newest first<br>View workout by ID<br>Update workout<br>Delete workout |
| FR-4 | Workout Search | Search by workout name keyword<br>Search by category<br>Search by date (YYYY-MM-DD) |
| FR-5 | AI Workout Recommendation | Enter age, fitness goal and experience level<br>Receive a personalised workout plan from Google Gemini |
| FR-6 | AI Fitness Insights | Enter total workouts, average duration and total calories<br>Receive a short AI insight with guidance |
| FR-7 | Access Control and Data Privacy | JWT protection on workout and AI routes |

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| | | Each user can access only their own workouts |
| FR-8 | Validation and Error Handling | Input validation with clear messages<br>Central JSON error responses for invalid IDs, duplicates and validation errors |

**Non-functional Requirements:**

Following are the non-functional requirements of the proposed solution.

| FR No. | Non-Functional Requirement | Description |
|---|---|---|
| NFR-1 | **Usability** | Consistent JSON responses ({ success, data / error }) with clear messages; Postman collection and README make the API easy to try and integrate. |
| NFR-2 | **Security** | Passwords hashed with bcrypt, JWT-based authentication, per-user data scoping, secrets kept in a git-ignored .env file, CORS enabled and all AI routes protected. |
| NFR-3 | **Reliability** | Central error handler, Mongoose schema validation, database connection check on start-up and graceful shutdown on unhandled promise rejections. |
| NFR-4 | **Performance** | Non-blocking Node.js I/O, lightweight queries sorted by date, and short AI prompts (2–3 paragraphs / sentences) using a fast Gemini Flash model. |
| NFR-5 | **Availability** | Stateless API that can run as several instances behind a load balancer, with a managed MongoDB Atlas cluster for the database. |
| NFR-6 | **Scalability** | Modular MVC structure and stateless authentication allow horizontal scaling; the AI service layer and MongoDB Atlas can scale independently. |